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Impact of hypoxic hepatitis on mortality in the intensive care unit

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The study was performed at the intensive care units 13H1, 13H3, and 13I2 of the Medical University of Vienna.

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Abstract *Purpose:* Hypoxic hepatitis (HH) is a form of hepatic injury following arterial hypoxemia, ischemia, and passive congestion of the liver. We investigated the incidence and the prognostic implications of HH in the medical intensive care unit (ICU). *Methods:* A total of 1,066 consecutive ICU admissions at three medical ICUs of a university hospital were included in this prospective cohort study. All patients were screened prospectively for the presence of HH according to established criteria. Independent risk factors of mortality in this cohort of critically ill patients were identified by a multivariate Poisson regression model. *Results:* A total of 118 admissions (11%) had HH during their ICU stay. These patients had different baseline characteristics, longer median ICU stay (8 vs. 6 days, $p < 0.001$), and decreased ICU survival (43 vs. 83%, $p < 0.001$). The crude mortality rate ratio of admissions with HH was 4.62 (95% CI 3.63–5.86, $p < 0.001$). Regression analysis demonstrated strong mortality risk for admissions with HH requiring vasopressor therapy (adjusted rate ratio 4.91; 95% CI 2.51–9.60, $p < 0.001$), whereas HH was not significantly associated with mortality in admissions without vasopressor therapy (adjusted rate ratio 1.79, 95% CI 0.52–6.23, $p = 0.359$). *Conclusions:* Hypoxic hepatitis (HH) occurs frequently in the medical ICU. The presence of HH is a strong risk factor for mortality in the ICU in patients requiring vasopressor therapy.

Keywords Hypoxic liver injury · Ischemic hepatitis · Shock liver · Mortality · Vasopressors · Epidemiology · Outcome

Abbreviations

ALT	Serum alanine transaminase
AST	Serum aspartate transaminase
CI	Confidence interval
ICU	Intensive care unit
IQR	Interquartile range
HH	Hypoxic hepatitis
RR	Rate ratio
SAPS	Simplified acute physiology score

Introduction

Hypoxic hepatitis (HH), also known as shock liver or ischemic hepatitis, is a diffuse hepatic injury following acute arterial hypoxemia, hypoperfusion with consecutive ischemia, and passive congestion of the liver. The disease is usually defined via acute, considerable, and reversible elevation of serum aminotransferase levels in the appropriate clinical setting. The reported incidence of this event ranges between 0.03% in the internal normal ward to 1.5% in the intensive care unit (ICU) [1, 2]. Recently, we reported that high international normalized ratio (INR), presence of septic shock, and high sequential organ failure assessment (SOFA) score were independent predictors of mortality in patients with HH [3]. Although it is reported that hyperbilirubinemia and other abnormal "liver function tests" on ICU admission may have a prognostic impact [4, 5], there are a lack of studies evaluating the influence of specific disease entities of acute hepatic injury—like HH—on outcome in critically ill patients. The main epidemiological data regarding this disease are more than 15 years old [2], or were obtained retrospectively [6] and may therefore not represent the current status of this disease.

Mortality in patients with HH is high, with a reported hospital mortality of about 50% [2]. In contrast to acute liver failure, the reported high mortality rate in patients with HH is usually attributed to the underlying condition that has led to the liver injury. Therefore, HH is considered to represent a complication in a broad spectrum of mainly cardiovascular diseases or sepsis rather than a single organ dysfunction [7]. A specific effect of HH on mortality unrelated to the severity of disease would therefore be surprising. The aim of this study was to examine the incidence of HH and its impact on mortality in patients admitted to three medical ICUs in a university hospital.

Materials and methods

Patients

The study was performed from March 2005 to December 2006 at three medical ICUs of the Medical University of Vienna after approval by the local Ethics Committee of the Medical University of Vienna, Austria. Patients admitted to the ICU were screened prospectively for the presence of HH.

Survival of a patient was assessed by examining the Austrian national death registry, contacting the patient or the attending physician, if necessary.

Data were prospectively entered into the ICU database on a daily basis. Preexisting diseases (cardiomyopathy, liver cirrhosis) were documented according to the medical history.

Liver cirrhosis was defined via histology or by a combination of characteristic clinical (ascites, caput medusae, spider angiomata, etc.), laboratory, and radiological findings (typical morphological changes of the liver, signs of portal hypertension etc., in ultrasonography or computed tomography scanning).

Cardiomyopathy was defined according to well-established criteria [8] and included several etiologies. Diagnosis was established via echocardiography or according to recent reports prior to ICU admission. Diseases contributing to the ICU stay (acute myocardial infarction, cardiopulmonary resuscitation, septic shock) were documented according to clinical findings during the ICU stay.

Hypoxic hepatitis

HH was defined according to well-established criteria [2, 9–12]: (a) clinical setting of cardiac, circulatory, or respiratory failure; (b) dramatic but transient increase in serum aminotransferase activity reaching at least 20-fold the upper limit of normal (normal ranges at the Medical University Hospital of Vienna, serum aspartate transaminase (AST) <35 U/L, serum alanine transaminase (ALT) <45 U/L); (c) exclusion of other putative causes of liver cell necrosis, particular viral or drug-induced hepatitis.

Liver biopsy was not required for the diagnosis of HH, in agreement with other studies showing that a histologic confirmation is unwarranted [2, 9, 11] and even inadvisable when the criteria listed above are met.

Patients were prospectively screened every day for the presence of HH. Every case of HH that occurred during the observation period at one of the three ICUs was included.

Statistical analysis

Data are presented as median and 25–75% interquartile range (IQR) or mean and standard deviation as appropriate. The effects of the independent variables on mortality rate are expressed as mortality rate ratios (RR) and 95% confidence intervals (CI).

For univariable analysis Mann–Whitney *U* test or chi-squared test were used as appropriate. The primary association tested was the influence of HH on survival with patient admission being the unit of analysis. We calculated stratum-specific rates and calculated Mantel–Haenszel rate ratios with 95% confidence intervals. The Mantel–Haenszel chi-squared test was used to test bivariate adjusted associations. Bivariate effect modification was tested by a chi-squared test.

A multivariable Poisson model was developed to adjust for potential confounders, which were defined a priori. We entered variables into the model if they were associated with survival and with HH, and if they were not considered to be on the causal pathway of the main association from conceptual reasoning. Linear assumptions and first-order interactions were also tested by using the likelihood ratio test. To allow for non-independence of patients with more than one readmission we calculated the presented confidence intervals based on robust standard errors.

As a sensitivity analysis to assess whether the model assumptions regarding readmissions were robust we calculated all data based on the individual patient as the unit of analysis.

We used MS Excel and Stata 8 for data management and calculations. A two-sided p value less 0.05 was generally considered statistically significant.

Results

Patients' characteristics

The three ICUs had 1,066 admissions during the observation period, of which 118 (11%) fulfilled the criteria of HH. Admissions with HH had a higher simplified acute physiology score (SAPS) 2 ($p < 0.001$) and SOFA score ($p < 0.001$), a significant longer stay at the ICU ($p < 0.001$), and significantly increased ICU mortality ($p < 0.001$) (Table 1). Twelve admissions (1%) without HH suffering from acute liver failure were treated at the ICUs during the study period. Liver cirrhosis was present in 78 ICU admissions. The underlying cause of cirrhosis was alcohol in 67%, viral hepatitis in 18%, and other causes in 15%. Mean model of end-stage liver disease (MELD) score was 25 ± 8. The Child–Pugh class was A in 3, B in 20, and C in 55 admissions. Admissions with cirrhosis represented 14% of the HH cohort and 6% of the non-HH cohort as illustrated in Table 1.

A total of 758 admissions required vasopressor therapy. The underlying causes necessitating vasopressor therapy are illustrated in the "Appendix". A total of 758 admissions received norepinephrine. The median dose on ICU admission was 0.2 (IQR 0.1–0.4) µg/kg/min, admissions with HH required a significantly higher norepinephrine dose than admissions without HH [median 0.42 (IQR 0.3–0.79) µg/kg/min vs. 0.18 (0.09–0.34) µg/kg/min, $p < 0.001$]. Seventy-six admissions received vasopressin additionally. Median vasopressin dose was 0.03 (IQR 0.03–0.07) units/min and did not differ significantly in admissions with [median 0.04 ((IQR 0.03–0.05) units/min] and without HH [median 0.03 (IQR 0.03–0.07) units/min] ($p = 0.537$). Dopamine was not administered and seven admissions received epinephrine.

Liver function parameters on admission to the ICU differed significantly in admissions with and without HH. Table 2 illustrates the detailed data including peak laboratory data in admissions with HH. The SOFA liver subset score was significantly and linearly associated with HH (p for trend ≤ 0.0001) in terms of an about 40% increase in the probability of having HH with increase in each unit SOFA liver subset score (OR 1.39, 95% CI 1.19–1.62; $p \leq 0.0001$). In detail, the SOFA liver subset score was

Table 1 Admissions characteristics

Parameter	All	Admissions without HH	Admissions who developed HH	p value
Number of admission	1,066	948	118	
Admissions' characteristics				
Age in years, median (IQR)	62 (51–72)	62 (50–72)	65.5 (56–73)	0.015
Male, *n* (%)	670 (62%)	590 (62%)	80 (68%)	0.238
SAPS 2, median (IQR)	48 (35–63)	47 (34–61)	64.5 (48–79)	<0.001
SOFA score, median (IQR)	9 (6–12)	9 (6–12)	11 (7–14)	<0.001
GCS, median (IQR)	13 (5–15)	13 (6–15)	11.5 (3–15)	0.016
Acute myocardial infarction, *n* (%)	134 (13%)	103 (11%)	31 (26%)	<0.001
Septic shock, *n* (%)	227 (21%)	190 (20%)	37 (31%)	0.005
Cardiopulmonary resuscitation, *n* (%)	197 (18%)	162 (17%)	35 (30%)	0.001
Liver cirrhosis, *n* (%)	78 (7%)	61 (6%)	17 (14%)	0.002
Cardiomyopathy, *n* (%)	181 (17%)	136 (14%)	45 (38%)	<0.001
Treatment				
Vasopressor, *n* (%)	758 (71%)	654 (69%)	104 (88%)	<0.001
Mechanical ventilation, *n* (%)	752 (71%)	655 (69%)	97 (82%)	0.003
Renal replacement therapy, *n* (%)	187 (18%)	150 (16%)	37 (31%)	<0.001
Outcome				
Length of ICU stay, days (IQR)	6 (3–12)	6 (3–11)	8 (5–17)	<0.001
ICU survival, *n* (%)	835 (78%)	784 (83%)	51 (43%)	<0.001

IQR interquartile range, *n* number, *SAPS* simplified acute physiology score, *SOFA* sequential organ failure assessment, *GCS* Glasgow coma scale, *ICU* intensive care unit

Table 2 Laboratory characteristics

Parameter	All admissions	Admissions without HH	Admissions who developed HH	*p* value
Laboratory characteristics on ICU admission				
Number of admissions	1,066	948	118	
AST (U/L), median (IQR)	49 (27–115)	44 (26–95)	291 (81–1,104)	<0.001
ALT (U/L), median (IQR)	31 (17–69)	28 (16–59)	140 (32–660)	<0.001
Bilirubin (mg/dL), median (IQR)	0.88 (0.55–1.63)	0.83 (0.53–1.48)	1.58 (0.83–2.57)	<0.001
Prothrombin time (%), median (IQR)	69 (48–85)	70 (51–87)	52 (33–71)	<0.001
Lactate (mmol/L), median (IQR)	1.5 (1–2.7)	1.4 (1–2.3)	3.2 (1.5–8.1)	<0.001

Parameter	Admissions with HH
Peak laboratory characteristics in admissions during the course of HH	
AST (U/L), median (IQR)	2,510 (1,317–6,732)
ALT (U/L), median (IQR)	1,366 (710–2,769)
Bilirubin (mg/dL), median (IQR)	2.6 (1.7–5.2)
Prothrombin time (%), median (IQR)	31 (17–44)
Lactate (mmol/L), median (IQR)	8 (4.4–13.5)

AST serum aspartate transaminase level, *ALT* serum alanine transaminase level, *HH* hypoxic hepatitis, *ICU* intensive care unit

0, 1, 2, 3, and 4 points in 645 (68.0%), 126 (13.3%), 123 (13.0%), 22 (2.3%), and 32 (3.4%) admissions without HH and in 52 (44.1%), 29 (24.6%), 26 (22.0%), 3 (2.5%), and 8 (6.8%) admissions with HH, respectively.

Univariate and bivariate analysis

Admissions' characteristics are illustrated in Table 1.

Median time of follow-up was 195 days (IQR 25–326 days) in the total cohort. Median survival differed significantly in admissions with HH versus admissions without HH (14 days vs. 219 days, $p < 0.001$). The overall 30- and 60-day mortality rates were 308 (29%) and 335 (31%), respectively, and differed significantly between admission with and without HH (30- and 60-day mortality rates were 63 and 66% in patients with HH versus 25 and 27% in patients without HH, $p \leq 0.001$). The figure in the "Appendix" illustrates the Kaplan–Meier curve of survival. The crude mortality rate ratio in admissions with HH was 4.62 (95% CI 3.63–5.86, $p < 0.001$).

The bivariate analysis examining the effect of HH in several strata on mortality is illustrated in Table 3. The presence of HH was significantly associated with increased mortality in all examined strata except in admissions that did not require vasopressor therapy. Several factors acted as effect modifiers (see Table 3, *p* value for heterogeneity). Note that the presence of liver cirrhosis was not an effect modifier in the bivariate analysis.

Multivariate analysis

We entered age, sex, SAPS 2 score, liver cirrhosis, cardiomyopathy, and renal replacement therapy as potential confounders of the association between mortality and HH into the model. As a consequence of vasopressor therapy acting as an effect modifier without being a potential confounder we built separate multivariable models for patients with and without vasopressor therapy. HH was a major independent risk factor for mortality (RR 4.91, 95% CI 2.51–9.60, $p < 0.001$) in admissions requiring vasopressor therapy. In contrast, HH was not significantly associated with mortality in admissions without vasopressor therapy (RR 1.79, 95% CI 0.52–6.23, $p = 0.359$). Age, SAPS 2 score, and sex were independent predictors for mortality in admissions not requiring vasopressor therapy during the ICU stay. Aside from HH, the presence of cardiomyopathy, presence of cirrhosis, necessity of renal replacement therapy, age, and a high SAPS 2 score were significant risk factors for mortality in admissions with vasopressor therapy. Table 4 presents the full Poisson models.

Sensitivity analysis using individual patient as the unit of analysis yielded virtually identical results.

Discussion

We prospectively investigated the incidence of HH and its impact on mortality in three medical ICUs of a university hospital. HH was found in 11% of ICU admissions and was a strong independent predictor of overall mortality in patients requiring vasopressor therapy.

Our data demonstrate a surprisingly high incidence of HH in the ICU in comparison to previous literature. This may be explained by several reasons. First, most of the previous studies were performed retrospectively and patients with HH may have been missed [6, 13]. Accordingly, Whitehead et al. [14] reported in a prospective study that more than one-third of all cases of notably raised aminotransferase levels were not noticed by the attending physician. Second, several previous

Table 3 Bivariate analysis (effect of hypoxic hepatitis on mortality including diverse strata)

Parameter	Rate ratio (95% CI)	*p* value for heterogeneity
Age		
18–50	3.59 (1.77–7.30)	
51–61	4.90 (3.01–7.98)	
62–71	5.69 (3.79–8.53)	
72–92	3.55 (2.28–5.54)	
Age-adjusted survival	4.48 (3.53–5.69)	0.408
SAPS 2		
2–34	5.42 (2.56–11.51)	
35–47	8.90 (4.76–16.64)	
48–62	2.77 (1.69–4.54)	
63–117	2.67 (1.91–3.75)	
SAPS 2-adjusted survival	3.20 (2.51–4.07)	0.002
Acute myocardial infarction		
No	4.53 (3.46–5.92)	
Yes	5.44 (3.06–9.66)	
AMI-adjusted survival	4.70 (3.69–6.00)	0.568
Cardiomyopathy		
No	4.97 (3.69–6.69)	
Yes	2.68 (1.76–4.10)	
Cardiomyopathy-adjusted survival	3.82 (2.99–4.88)	0.017
CPR		
No	4.78 (3.61–6.31)	
Yes	3.62 (2.27–5.77)	
CPR-adjusted survival	4.39 (3.45–5.58)	0.315
Septic shock		
No	3.85 (2.84–5.22)	
Yes	7.27 (4.92–10.75)	
Septic shock-adjusted survival	4.75 (3.74–6.03)	0.011
Cirrhosis		
No	4.52 (3.48–5.86)	
Yes	3.59 (1.97–6.56)	
Cirrhosis-adjusted survival	4.33 (3.40–5.50)	0.493
RRT		
No	4.94 (3.69–6.61)	
Yes	2.61 (1.71–3.96)	
RRT-adjusted survival	3.83 (3.01–4.86)	0.013
Mechanical ventilation		
No	5.38 (3.10–9.34)	
Yes	4.17 (3.19–5.44)	
Mechanical ventilation-adjusted survival	4.34 (3.42–5.52)	0.411
Vasopressor therapy		
No	1.74 (0.76–3.99)	
Yes	5.09 (3.95–6.57)	
Vasopressor therapy-adjusted survival	4.48 (3.52–5.69)	0.012

CI confidence interval, *SAPS* simplified acute physiology score, *CPR* cardiopulmonary resuscitation, *RRT* renal replacement therapy

studies were performed in the 1980s and 1990s [2, 11]. There are several reasons to suggest that HH may nowadays be more common: the increasing age and comorbidities of the patients [15], and the increasing number of sepsis, severity of disease, and organ failure [16–18]. Therefore, the incidence of HH may have also increased over time.

According to our experience, HH is by far the most frequent cause of acute liver injury in the medical ICU: only 12 (1%) admissions with acute liver failure were treated during the observation period in contrast to 118 (11%) admissions with HH. This in concordance with previous laboratory studies that identified HH as the most common cause of notably raised aminotransferase levels in hospitals [14, 19]. In summary, our data indicate that critical care physicians should be aware of this severe disease as they will be frequently confronted with HH in their daily practice.

Previous reports suggested that mortality in patients with HH is frequently attributed to the underlying condition that has led to the liver injury [6, 7, 20]. However, this seems to be only true in patients not requiring vasopressor therapy according to our experience. HH per se is a strong independent risk factor for mortality in critically ill patients requiring vasopressor therapy. Consequently, HH should be recognized as a major independent prognostic factor in hemodynamically unstable ICU patients with shock state.

Table 4 Multivariable Poisson regression model for risk factors of mortality

	Admissions without vasopressor therapy		Admissions with vasopressor therapy	
Number of patients	308 (29%)		758 (71%)	
Parameter	Rate ratio (95% CI)	*p* value	Rate ratio (95% CI)	*p* value
Hypoxic hepatitis	1.79 (0.52–6.23)	0.359	4.91 (2.51–9.60)	<0.001
Cardiomyopathy	1.18 (0.56–2.49)	0.672	2.32 (1.48–3.64)	<0.001
HH × CMP[a]	1.31 (0.16–10.78)	0.803	0.49 (0.18–1.31)	0.156
Renal replacement therapy	1.19 (0.48–2.95)	0.803	3.73 (2.46–5.66)	<0.001
HH × RRT[b]	0.16^{-6} (0.68^{-8}–0.36^{-5})	<0.001	0.35 (0.13–0.96)	0.041
Cirrhosis	1.73 (0.54–5.48)	0.702	3.59 (1.69–7.65)	0.001
Sex (male vs. female)	2.26 (1.19–4.29)	0.013	1.06 (0.74–1.52)	0.759
SAPS 2[c]				
Quartile 2 versus quartile 1	1.69 (0.86–3.30)	0.127	0.71 (0.39–1.30)	0.269
Quartile 3 versus quartile 1	2.19 (1.05–4.55)	0.036	1.33 (0.78–2.28)	0.295
Quartile 4 versus quartile 1	5.19 (1.86–14.48)	0.002	2.27 (1.34–3.85)	0.002
Age[d]				
Quartile 2 versus quartile 1	3.76 (1.54–9.17)	0.004	2.37 (1.17–4.80)	0.017
Quartile 3 versus quartile 1	2.45 (0.89–6.74)	0.083	2.66 (1.33–5.32)	0.006
Quartile 4 versus quartile 1	2.99 (1.16–7.68)	0.023	3.46 (1.68–7.11)	0.001

HH hypoxic hepatitis, *CMP* cardiomyopathy, *RRT* renal replacement therapy, *SAPS* simplified acute physiology score

[a] Interaction term for HH × CMP

[b] Interaction term for HH × RRT

[c] SAPS 2 quartile 1, 2–34; quartile 2, 35–47; quartile 3, 48–62; quartile 4, 63–117

[d] Age quartile 1, 18–50 years; quartile 2, 51–61 years; quartile 3, 62–71 years; quartile 4, 72–92 years

HH constitutes a heterogeneous condition and we found three significant interactions on the effect of HH on mortality, indicating that the main effect varies across several groups. There are several ways of dealing with interactions, and we have chosen two methods in this report. For the clinically most unambiguous variable "vasopressor therapy" we present a separate model each for patients with vasopressor and for patients without vasopressor. For the other two interacting variables "cardiomyopathy" and "renal replacement therapy" we presented interaction effects along with the usual effects within those two models, because otherwise we would have ended up with eight different models. However, exposed risk can still be readily calculated by multiplying the interaction effects with other effects as appropriate in plain multivariable models.

Why does HH have such a central prognostic role in patients receiving vasopressor therapy? The principal response of the hepatic vascular bed to catecholamines is vasoconstriction [21, 22]. Norepinephrine and epinephrine divert blood flow away from the mesenteric circulation and decrease microcirculatory blood flow in the gastrointestinal tract despite increased perfusion pressure and increased systemic blood flow in experimental sepsis [23]. These may be a consequence of impaired hepatic blood flow regulation and prolongation of vasopressor effects due to reduced hepatocyte metabolism [24]. Furthermore, experimental data suggest that catecholamines may deteriorate hepatocellular function by induction of an inflammatory response syndrome [25]. Whether vasopressor therapy may aggravate the hepatic damage and impede recovery of the acutely injured liver in patients with HH should be clarified by future studies. On the other hand, hepatic impairment can also trigger progression of organ dysfunction per se [26]. HH can be associated with hepatopulmonary syndrome and can aggravate hypoxemia [20]. Experimental acute ischemia reperfusion injury may contribute to acute myocardial damage, acute lung, and kidney injury [27–29]. Furthermore, the liver seems to modulate the systemic inflammatory response syndrome via synthesis and release of acute-phase proteins and cytokines [30, 31]. Thus, several causalities may explain the strong prognostic impact of HH, especially in patients requiring vasopressor therapy.

What are the implications of this study? On the one hand, clinicians should be aware of the prognostic importance of HH. They should try to recognize the occurrence of HH as early as possible. As aminotransferase levels usually rise 12–24 h after the initiating event that has led to the acute liver injury, several methods may help to detect acute hepatic injury earlier [7, 26]. Indocyanine green plasma disappearance rate (ICG-PDR) is a method of assessing the liver function in acute and chronic liver injury. ICG-PDR illustrates the current hepatic function without delay. Therefore it seems to be a useful diagnostic method in critically ill patients [26]. However, there are still a lack of clinical data proving this assumption. Apart from its diagnostic properties, ICG-PDR may be useful in monitoring the course of the disease. Furthermore, it is a good predictor of survival in critically ill patients and may be a helpful prognostic tool in patients with HH [26, 32, 33]. Other methods like the MEGX test

or assessment of hepatic blood flow by galactose clearance may be useful alternatives [34, 35]. Several authors could demonstrate a significant reduction of hepatic blood flow in patients with HH following heart failure [6, 35]. In cases of HH, physicians should try to stabilize the hepatic function as early as possible to avoid possible complications of the hepatic injury which may worsen multiorgan failure [3, 26]. As no specific therapies improving the hepatic function in patients with HH are currently established, the main effort of clinicians should focus on identifying and reversing the basic HH-causing disease. On the other hand, researchers should draw their attention to therapeutic options in ameliorating the hepatic injury in patients with HH. Low-dose dopamine may impede the development of hypoxic liver injury [36]. However, dopamine was not used in our cohort. Substances influencing the renin angiotensin system may provide possible options as indicated by experimental ischemia reperfusion models [37, 38]. There is also evidence that nitrite may mediate cytoprotection in hepatic ischemia reperfusion injury [39, 40]. However, to date there are a lack of interventional studies in patients with HH.

Interestingly, we could not observe any difference on HH-related mortality between admissions with and without liver cirrhosis. This may be a consequence of the small number of admissions with cirrhosis and HH in our study. Future studies should investigate the impact of HH in patients with cirrhosis in larger cohorts.

There are limitations of our study. First, several subgroups are present in our population. This heterogeneity ends up in some strata in which hardly any interference can be made (Table 3). Second, the study was performed at three medical ICUs of a university hospital. Accordingly, the conclusions drawn form this study might not necessarily be generalized to surgical wards or tertiary care hospital wards. However, there seems to be no major potential for observer bias, as the risk factor classification was independent of the outcome. Furthermore, classification of mortality is a robust outcome parameter even in cases of unblinded assessment.

In conclusion, HH occurred in 11% of all admissions to three medical ICUs of a university hospital. More than half of the patients with HH died at the ICU. HH is a strong independent risk factor for mortality in critically ill patients requiring vasopressor therapy. The high incidence and the strong prognostic impact on ICU survival should encourage research to recognize the occurrence of HH as early as possible, to evaluate therapeutic options in HH, and to ameliorate the course of HH.

Conflict of interest The authors of have nothing to disclose.

Appendix

See Fig. 1 and Table 5

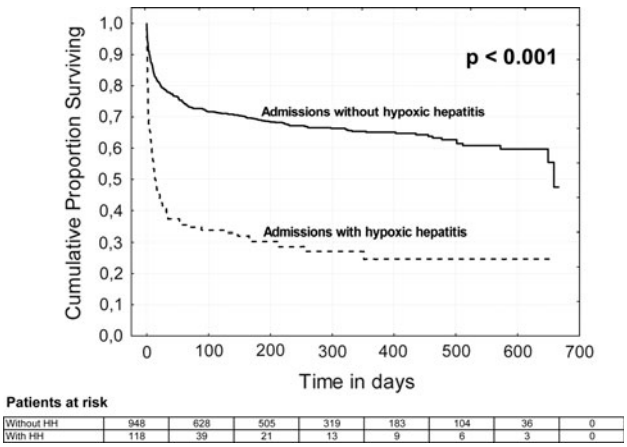


Fig. 1 Cumulative survival rate of 118 admissions with hypoxic hepatitis and 948 admissions without hypoxic hepatitis. Survival was significantly higher in patients without hypoxic hepatitis (log rank test)

Table 5 Main underlying admission diagnosis stratified according to the presence or absence of vasopressor therapy

Diagnosis	Admissions with vasopressor therapy	Admission without vasopressor therapy	*p* value
Number of admissions	758	308	
Acute myocardial infarction, *n* (%)	115 (15)	19 (6)	<0.001
Cardiopulmonary resuscitation, *n* (%)	102 (13)	16 (5)	<0.001
Decompensated cardiomyopathy, *n* (%)	68 (9)	65 (21)	<0.001
Valvular heart disease, *n* (%)	22 (3)	20 (6)	0.009
Pulmonary embolism, *n* (%)	16 (2)	25 (8)	<0.001
Septic shock, *n* (%)	227 (30)		
Hemorrhagic shock, *n* (%)	28 (4)		
Pneumonia without shock, *n* (%)		50 (16)	
Surgery, *n* (%)	78 (10)	25 (8)	0.305
Neurological disease, *n* (%)	41 (5)	32 (10)	0.007
Acute liver failure, *n* (%)	5 (1)	7 (2)	0.047
Oncology, *n* (%)	10 (1)	12 (4)	0.015
Other, *n* (%)	56 (7)	37 (12)	0.02

104 admissions with hypoxic hepatitis (HH) had vasopressor therapy, 14 admissions with HH required no vasopressor therapy

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